

Leveraged Multi-Strategy Ensemble Trading System

1 Methodology Overview

The **CombinedStrategy** is an ensemble quantitative model designed to provide diversified exposure through three distinct sub-strategies: Defensive Adaptive Asset Allocation (DAA), Kalman-based Pairs Trading, and a static Gold anchor. The model utilizes a hierarchical leverage structure to enhance returns while maintaining defensive guardrails.

2 Core Components

2.1 Defensive Adaptive Asset Allocation (DAA)

The DAA component serves as the primary growth engine. It utilizes a "Canary" mechanism to detect regime shifts.

- **Canary Assets:** VWO (Emerging Markets) and BND (Total Bond Market). Momentum Scoring: Assets are ranked using a weighted momentum formula:

$$12 \left(\frac{P_t}{P_{t-22}} - 1 \right) + 4 \left(\frac{P_t}{P_{t-63}} - 1 \right) + 2 \left(\frac{P_t}{P_{t-126}} - 1 \right) + \left(\frac{P_t}{P_{t-252}} - 1 \right) \quad (1)$$

- **Defensive Logic:** If all canary assets exhibit negative momentum, 100% of the DAA slice moves to IEF (Intermediate Treasuries).

2.2 Statistical Arbitrage (Pairs Trading)

The strategy identifies price dislocations between correlated assets using a state-space model.

- **Kalman Filter:** Dynamically estimates the hedge ratio β and intercept α for the linear relationship $y_t = \beta x_t + \alpha + \epsilon_t$.
- **Z-Score Entry:** Trades are signaled when the residual ϵ_t deviates significantly from its moving standard deviation:

$$Z = \frac{\text{spread} - \mu_{\text{spread}}}{\sigma_{\text{spread}}} \quad (2)$$

3 Portfolio Construction

The portfolio is rebalanced monthly for DAA and daily for Gold. Leverage is applied at the sub-strategy level.

Component	Allocation	Leverage	Total Exposure
Gold Anchor	20%	1.00x	20%
DAA Momentum	55%	1.45x	79.75%
Pairs Trading	25%	1.95x	48.75%